

FETAL CRANIAL SEGMENTATION IN 2D ULTRASOUND IMAGES USING SHAPE PROPERTIES OF PIXEL CLUSTERS

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ABSTRACT

The detection of cranial dysmorphisms during pregnancy is achieved by assessing the cranial shape from 2D ultrasound images of the fetal head. As such, several algorithms have been presented to automate this task due to the fact that segmentation of the fetal cranium from ultrasound images is a central problem in obstetric care which is complicated by fuzzy boundaries and variability in fetal position and head shape. In this paper, we introduce a machine learning framework that employs a novel feature set which incorporates local statistics and shape information about pixel clusters (or superpixels) within an image, and evaluate the performance of the feature set in the task of segmenting the cranial pixels in an ultrasound image using a random forest classifier. Our experiments show that the features derived from the shapes of the pixel groupings outperform powerful features such as Haar features and achieved a **97.22%** segmentation accuracy when applied to the task of fetal cranial segmentation in ultrasound images.

Index Terms— Fetus, Ultrasound imaging, Skull, Cranium, Random forests, Superpixels

1. INTRODUCTION

In obstetric care, ultrasound images of the fetal head are collected in order to obtain cranial biometry measurements (i.e. head circumference and diameters) and for the diagnosis of craniofacial dysmorphology. Cranial malformations have been linked to chromosome abnormality and in excess of 150 syndromes with clinical implications [1]. Thus, accurate segmentation and extraction of the cranial pixels is paramount to the assessment of fetal health. In the literature, the proposed algorithms are based on morphological operators [2, 3] Hough transform ellipse extraction [4, 5], active contour models [6, 7], and deformable models [8]. We propose an automated machine learning approach for segmenting the fetal cranium from 2D ultrasound images using localized shape information from pixel clusters (or *superpixels*). Superpixels provide local statistics about the pixels in a given neighbourhood and reduce image redundancy by grouping

pixels belonging to the same object. Superpixels have been successfully applied to the segmentation of mitochondria in electron microscopy images, in a framework that operated on the superpixels themselves and their respective shape features for classification [9]. The quality of such a segmentation is limited by the resolution and boundary adherence of the superpixel oversegmentation, as it does not consider the individual pixels in the original image. In this paper, we propose the use of individual pixel information in conjunction with superpixel-derived shape information as an element in the feature vector of each pixel in the image. We compare the performance of our proposed feature to powerful features that inform on local context information (e.g. Haar) as well as 'weak' intensity and positional features.

2. METHODS

2.1. Pixel Cluster Creation

In this work, we constructed the 2D superpixel lattice using the *simple linear iterative clustering* (SLIC) algorithm [10]. The algorithm assigns each pixel in the image to the nearest cluster center whose search area overlaps with the particular pixel. For an image of N pixels, the algorithm takes an input of the desired superpixel number, K , such that the approximate superpixel size is N/K and the cluster sampling length is determined as $S = \sqrt{N/K}$. Pixels are associated to a cluster center through the minimization of a modified Euclidean distance metric in every $2S \times 2S$ region of $\mathbf{I}(x, y)$

$$\delta_{jk} = \sqrt{\frac{(i_k - i_j)^2}{m^2} + \frac{(x_k - x_j)^2 + (y_k - y_j)^2}{S^2}}$$

where i_j and (x_j, y_j) are the pixel intensity and spatial coordinates of pixel j , respectively; i_k and (x_k, y_k) are those pertaining to the cluster center k ; m regulates the compactness of the superpixel. Lower values of m yield less compact superpixels that closely adhere to the image boundaries, and higher values result in more compact pixels. The cluster center position is then recomputed by averaging the ixy vector of all the pixels belonging to each cluster. The algorithm then proceeds iteratively, associating pixels to cluster and updating the cluster centers until convergence. We found that SLIC superpixel oversegmentation adapted well to our speckled ultrasound data, and was able to visually discriminate between

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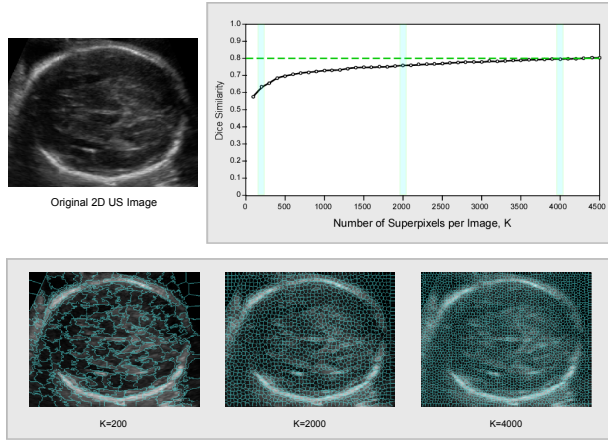


Fig. 1. Dice similarity coefficient with respect to the number of superpixels (K) Higher superpixel numbers yield greater adherence to the object boundaries and reduced oversegmentation. [For visualization of the segmentation scales, the reader is referred to a zoomed in version of the electronic document.]

different structures within the fetal head. Figure 1 shows the Dice similarity values for a range of $K = \{100, \dots, 4500\}$. For our multiscale cluster shape-based segmentation, we selected image cluster representations of the ultrasound image for multiple scales $K = [200 \ 2000 \ 4000]$, as they are in the different parts of the curve.

2.2. Feature Extraction

The feature vector of the pixel in the original image is comprised of reportedly powerful feature sets such as unary, binary, positional, and Haar features (Fig. 2). In addition, each pixel i in the original image $\mathbf{I}$ is associated to clusters of a different scales $k_i \in [200 \ 2000 \ 4000]$. The new shape feature set, $\mathbf{f}_i^{\text{Shape}}$, is extracted from each cluster k_i and added to the feature vector of each pixel.

$$\mathbf{f}_i = [\mathbf{f}_i^{\text{Unary}}, \mathbf{f}_i^{\text{Binary}}, \mathbf{f}_i^{\text{X,YPos}}, \mathbf{f}_i^{\text{Haar}}, \mathbf{f}_i^{\text{Shape}}]^\top$$

2.3. The Shape Feature Set, $\mathbf{f}_i^{\text{Shape}}$

The shape feature set comprises of six distinct features (Fig. 2). Each feature captures the local shape information of pixels that would belong to the same object in the image by exploiting the fact that cluster membership is defined by high similarity of pixel intensities and spatial proximity. Thus, the goal of the feature set is to consider the pixels as hierarchical entities by incorporating structural information from the pixel clusters of different scales to which the pixel belongs.

Given a pixel p_i , the shape features are computed on the pixel cluster k_i of the superpixel lattice of scale K , and these

are inserted into the feature vector of the pixel, $\mathbf{f}_i$. For a pixel cluster k_i , we construct the shape features by acquiring the pixel intensities and coordinate locations and computing the cluster shape descriptor for cluster k_i . The cluster shape descriptor for an image $\mathbf{I}$ of superpixel lattice scale K is given by

$$\mathbf{f}_i^{\text{Shape}}(\mathbf{I}, K) = [f^{[\mu\text{int}]}, f^{[\text{area}]}, f^{[\text{maj}]}, f^{[\text{min}]}, f^{[\text{ecc}]}, f^{[\text{circ}]}]^\top$$

where the individual features are shown in figure 2.2 and their details are shown on table 1.

Table 1. Description of Shape Features

Shape Feature	Equation	Description
$f^{[\mu\text{int}]}$	$\frac{1}{n} \sum_{i=0}^n \mathbf{I}_n$	The mean intensity of the pixels belonging to cluster k_i consisting of n pixels
$f^{[\text{area}]}$	n	Number of pixels in the cluster
$f^{[\text{maj}]}$	λ_2	Major axis length of an ellipse that best represents the cluster
$f^{[\text{min}]}$	λ_1	Minor axis length of an ellipse that best represents the cluster
$f^{[\text{ecc}]}$	$\sqrt{1 - \frac{\lambda_2}{\lambda_1}}$	Eccentricity of an ellipse that best represents the cluster
$f^{[\text{circ}]}$	$\frac{\text{Perimeter}^2}{2\pi n}$	The circularity (or roundness) of the cluster

The feature vector was obtained for each pixel in the training images and passed into the random forest classifier for training.

3. RANDOM FOREST CLASSIFIER

A random forest is an ensemble of trees, T , trained from a set of training examples, I , each defined by the original image and the class labels, c , pertaining to each pixel [11]. Each of the trees is trained separately on a random subset $I' \subseteq I$. During the training stage, the subset traverses through the tree, recursively being split into I'_q at each internal tree node q into left and right data subsets, I'_{ql} and I'_{qr} respectively, according to the threshold t of a selected splitting feature, f . The splitting feature is selected at each node in a tree from the feature vector $\mathbf{f}_i$ pertaining to pixel i in the training set, in order to yield maximal information gain (in this case, Shannon entropy). Each leaf node is then associated with learned probability distributions $P(c|k)$ of all the class labels c . The training continues recursively until the maximum tree depth D has been reached, or if the information gain is below a given threshold. This process is repeated for each tree in the

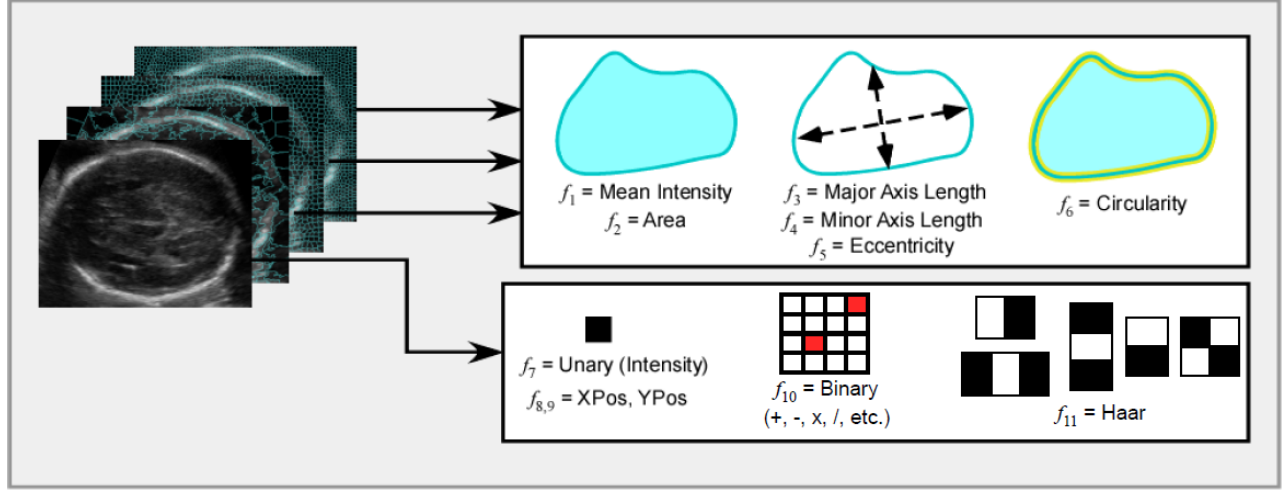


Fig. 2. Feature extraction from intensity and multiscale superpixel lattices.

forest, and the forest classification is achieved by averaging the class distributions over the leaf nodes $l_i, \dots, l_T \in L$,

$$P(c|L) = \frac{1}{T} \sum_{t=1}^T P(c|l_t)$$

4. RESULTS

4.1. Implementation Details

Our algorithm was trained on 60 randomly-selected 2D ultrasound images of the fetal head of fetuses from 25 to 34 weeks of gestation collected from the Intergrowth-21st database. The images had a resolution of 0.2mm and the dimensions ranged from 380×450 to 500×680 pixels such that the sizes of the pixel clusters ranged from approximately $N \sim [1270, 127, 63]$ pixels for $K = [200, 2000, 4000]$, respectively. The validation set consisted of images from 10 fetuses from 28 to 34 gestational weeks. The proposed algorithm used a single random forest for classification consisting of $T = 20$ trees with a maximum depth of 15 and 120 features were sampled at each node from which only the most discriminating feature was selected. The algorithm was implemented in C++ and training lasted ~ 1.4 min per tree (3.30GHz quad-core, 12GB RAM). Two separate trees were trained for comparison: one tree excluded f_i^{Shape} from the feature vector, whereas the other included our proposed feature set.

4.2. Segmentation Performance

Figures 3 displays the feature sets selected at each tree depth averaged across all 20 trees for a tree excluding f_i^{Shape} (Fig. 3a) and for a forest including the shape features (Fig. 3b). The results show that the inclusion of f_i^{Shape} in the feature vector reduces the frequency at which f_i^{Haar} is selected. That is, from a depth of 4 f_i^{Shape} is preferentially selected over f_i^{Haar} meaning that it has a higher discriminative power ($\geq 4.07\%$).

We also found that the most discriminative shape features are: a) $f^{[\mu_{\text{int}}]}$, whose value will be proportional to the brightness of the cranial pixels, b) $f^{[\text{area}]}$ which captures the number of adjacent pixels belonging to the cranium, and c) $f^{[\text{maj}]}$ which informs on how elongated the clusters are. The least powerful feature is $f^{[\text{ecc}]}$ possibly because it incorporates both $f^{[\text{maj}]}$ and $f^{[\text{min}]}$ into a single metric, so it is not preferentially selected when its constituent features are available.

In order to quantify the cranial segmentation accuracy, we display the Dice similarity results on table 2 for a tree depth of 6 (highlighted in blue in figure 3)—a depth at which the forest is not overfitting. The segmentation accuracy increased by **3.67%** with the inclusion of f_i^{Shape} in the feature vector. Representative segmentation results are shown in figure 4.

Table 2. Segmentation Performance of Feature Sets

Feature Set	Precision (%)	Recall (%)	Accuracy (%)
Intensity	99.52	93.80	93.55
Intensity + Shape	99.01	98.11	97.22

5. CONCLUSION

We have presented a framework which segments the cranial pixels of 2D ultrasound images using pixel-based information and shape statistics of pixel clusters in the feature vector of each image pixel. A random classification forest was used to obtain the probability maps of the pixels belonging to the fetal cranium. Our results show high segmentation accuracy and further developments may motivate the use of our framework for the assessment of *in utero* fetal cranial morphology.

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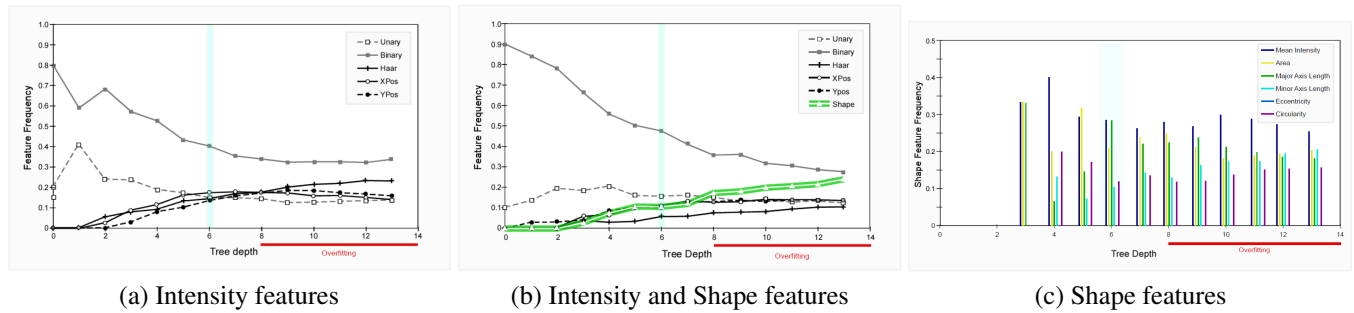


Fig. 3. Random feature selection. Selection of features at different tree depths for a random forest trained with only intensity features (a), intensity and shape features (b), and a histogram of the shape features selected at each depth (c) corresponding to the forest trained in (b).

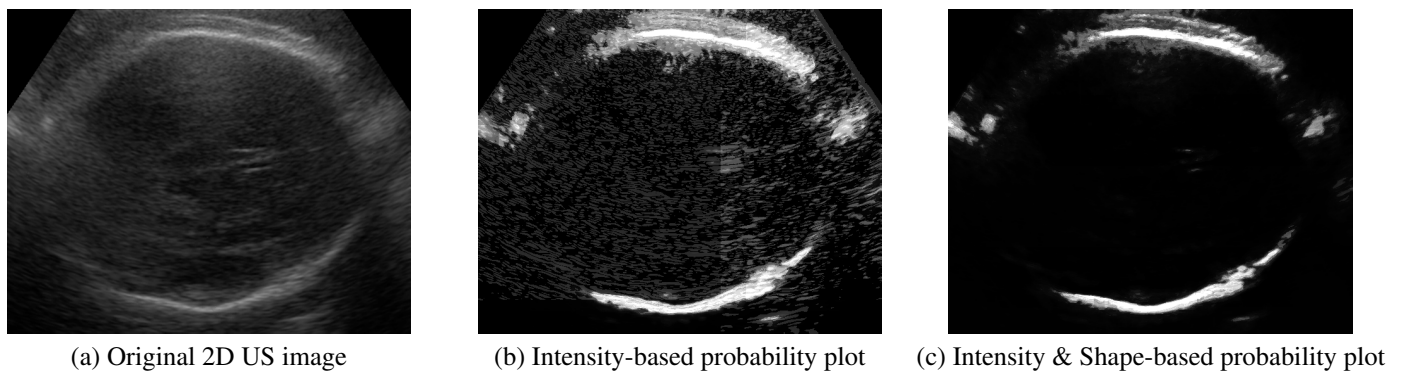


Fig. 4. Probability estimation plots. The probability plots computed via random forest classification on the original 2D ultrasound image (a) for an intensity-based forest (b) and an intensity+shape based forest (c) at tree depth=6. Higher pixel intensities correspond to a higher likelihood of a pixel belonging to the fetal cranium.

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